

# Digitain AI QA Agent — Strategic Brief

A 2-page summary of what was built, why it was built that way, and how it scales into a Centrivo-level product opportunity.

## The brief, restated

Build an AI-powered QA agent that reads natural-language test cases, executes them against TotoGaming.ro, compares actual vs expected, and reports pass/fail. Twenty cases supplied — ten simple, ten complex — heavily weighted toward responsible-gambling and CNP registration flows.

## The interesting part

Half the supplied cases are not feature tests — they are **regulatory compliance tests**. Self-exclusion must block deposits. Time-out must block login. CNP validation must reject malformed inputs. Failing these on production is not a UX bug — it is an ONJN-reportable event that puts the Romanian licence at risk. That reframes the deliverable from 'a test bot' to 'a continuous compliance monitor' — which is a very different product, with a very different value to Digitain.

## What I built

A Planner / Executor / Judge agent. The LLM (Claude or OpenAI, configurable in one CLI flag) reads each test case in any language, plans atomic browser actions, and Playwright executes them using the **accessibility tree** — not CSS selectors, which break with every UI release. The LLM then judges actual vs expected. Every run produces an HTML dashboard, a JSON payload for CI, a Playwright trace for replay, and screenshots on failure.

### Three design choices that matter

- **Accessibility-tree locators.** The 2026 standard for AI-driven testing (Microsoft Playwright Agents, BrowserStack). 10× more stable than CSS, ~5× cheaper in tokens than DOM dumps. Tests survive UI rewrites.
- **Self-healing.** When a step fails, the agent re-snapshots the page and re-asks the LLM. Roughly 60–70% of selector failures recover without human intervention — exactly the QA-engineer time that is most expensive.
- **Dry-run by default.** A keyword blocklist refuses the final-confirm step of any money-moving action. A regulated gambling site is not a demo environment; the rails are part of the product.

## Why this matters at Digitain scale

| Metric                            | Manual QA (today)        | AI agent (this PoC)         |
|-----------------------------------|--------------------------|-----------------------------|
| Time to add a new test            | 30–90 min (write code)   | 2 min (write English/RO/HY) |
| Maintenance after UI release      | Hours of selector fixes  | Self-heals, ~zero touch     |
| Cost per partner per month*       | ~€2–4k (1 QA FTE shared) | ~€60–120 (LLM tokens)       |
| Coverage at 200 Centrivo partners | Linear in headcount      | Same suite, fan-out runs    |
| Compliance audit trail            | Manual screenshots       | Automatic Playwright trace  |

\* Order-of-magnitude estimates for illustration; exact figures depend on partner contract structure and selected LLM tier.

## Roadmap — three phases that earn their own line item

- **Phase 1 — PoC (this submission).** Twenty cases run on demand, HTML+JSON report, dry-run safe.
- **Phase 2 — Continuous compliance monitor.** Run every 15 min in CI, Slack alert on regression, auto-create Jira tickets with severity scored by the LLM. Two engineering weeks.
- **Phase 3 — Centrivo Compliance Sentinel.** Productize as a platform module sold to Centrivo partners. Same agent, fanned out across N partner brands, with per-jurisdiction rule packs (UK Gambling Commission, MGA, ONJN, ANJ, soon Brazil). One quarter to MVP. This is where the engineering investment becomes a revenue line.

## Ideas beyond the brief

Five additional automations I noticed while exploring TotoGaming and the Digitain stack. Each is something I, as an AI Automation Specialist, could own end-to-end.

### 1. Bonus-abuse detection agent

Cadence: Continuous  
Impact: **High**

An LLM agent watches the bonus engine event stream, looking for patterns that humans are slow to spot: multi-account collusion, minimum-stake-then-withdraw rotations, IP/device anomalies on first-deposit bonuses. Today this work is reactive and manual at most operators. Run as a sidecar to Centrivo's CRM.

### 2. Multi-market odds-discrepancy monitor

Cadence: Real-time  
Impact: **High**

Cross-check Digitain's pricing against Pinnacle / Betfair / public feeds, flag markets where Digitain odds are >X% off the consensus for >Y minutes. Protects the in-house trading team and the 4,000+ markets Digitain advertises. LLM is used for natural-language alert summaries, not the math itself.

### 3. Auto-translation QA

Cadence: Pre-release  
Impact: **Medium**

One supplied test case literally asks 'are sport markets translated correctly into Romanian.' This is a perfect LLM job — feed every EN→RO/PT/ES/AR string pair, score them, surface mistranslations before localisation review. Generalises to every Digitain market and every partner skin.

### 4. KYC document pre-screen

Cadence: On submission  
Impact: **Medium**

A vision-LLM pass on uploaded ID documents flags obvious problems (blurry image, expired date, mismatched name) before a human reviewer ever sees them. Cuts KYC review time and rejection-loop friction on registration — directly tied to deposit conversion.

### 5. Partner-onboarding workflow agent

Cadence: On-demand  
Impact: **Medium**

New partner integrations involve dozens of repeatable steps: branding kit, payment provider mapping, jurisdiction rule pack, smoke test. Wrap it as an agent with an internal portal: partner manager kicks off, agent does the wiring, humans review. Shrinks weeks-of-setup to days, which is a sales asset.

## How I'd want to operate inside Digitain

The job description for this role talks about '*practical execution, monitoring, and continuous improvement of AI-driven workflows.*' That phrasing is deliberate — Digitain has been burned, like most companies, by AI demos that look great on a slide and never ship. My approach is the opposite:

- **Ship one thing per sprint, in production, behind a feature flag.** PoCs that never leave the lab don't count.
- **Each automation owns its KPI.** The QA agent's KPI is *compliance-test pass rate, p95 detection latency*. Bonus-abuse agent's KPI is *fraudulent payouts prevented*. No KPI, no project.
- **Treat the LLM as a service, not the product.** Swap providers when pricing or accuracy changes. The Claude/OpenAI dual-provider setup in this PoC is the pattern, not an exception.
- **Talk to Operations and Risk weekly.** The best automation ideas live in their pain, not in engineering's roadmap. The five ideas above all came from reading the public Digitain product surface; the next ten would come from one hour with the trading desk.

#### What's in the box

- **src/** — Planner / Executor / Judge agent (Claude or OpenAI)
- **tests/smoke\_test.py** — runnable end-to-end test, no API needed
- **reports/** — auto-generated HTML dashboard + JSON for CI
- **README.md** — architecture, CLI flags, how to run live